Matt Farzaneh

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Education

Western University — London, ON
B.Sc. Honors Computer Science, Minor in Software Engineering — *Expected April 2028*

Experience

Analyst Intern, Strategic Initiatives (Software Development) — May 2026 – Present
University Pension Plan Ontario — *Toronto, ON*

- Saved hours of manual work each reporting cycle by building **Python/PostgreSQL ETL pipelines** integrating Bloomberg, Albourne, and DataHub to automate currency conversion and reporting.
- Deployed a **Docker**-containerized **Streamlit** application on **GCP** for quarterly investment manager reviews.
- Built public equity and global macro agents with **Python, Google ADK, and Gemini Enterprise** to support investment manager analysis, with prompt-level safeguards against unsupported claims.
- Hardened the agents through **input metadata validation, file-path restrictions, credential protection, and output-token limits** to address data exposure and excessive usage risks.

Co-Founder / Developer — June 2025 – May 2026
After Image Inc — *Toronto, ON*

- Launched a production **B2B SaaS backend** using **Node.js on AWS**, with role-based access control separating tenant and administrator permissions.
- Maintained responsive tenant-scoped reporting as data grew by designing **MySQL schemas and REST APIs**, adding indexes, and tuning queries.
- Automated tenant billing and invoicing by integrating **Stripe** into a multi-tenant analytics and reporting dashboard.

Data Engineer & Business Analyst — May 2024 – Sept 2025
K&D Consulting — *Toronto, ON*

- Cut manual Excel processing by **95%** and developer intervention by **90%+** with a Python automation framework driven by YAML pipeline configurations.
- Saved **40+ hours of weekly reporting work** by automating **500+ monthly financial workbook integrations** while preserving data integrity.

Projects

Nasdaq ITCH Market Data Engine | *C++17, Python, CMake* | Code | Live — August 2026

- Achieved **43M messages/sec with zero allocations/message** in an in-memory replay benchmark by building a C++17 packet parser and order book with preallocated pools and open-addressed indexing.
- Validated **14.27M events with zero mismatches across 3.77M checkpoints** against Databento's 10-level reference book, identifying and fixing a duplicate-execution bug.
- Automated correctness and performance regression checks on **Linux and Windows with GitHub Actions**, including C++/Python tests, differential replay, and a zero-allocation check.

Urbanova (Hack the Valley 11) | *TypeScript, Python, PostgreSQL* | Code | Live — October 2026

- Built an **Express/TypeScript backend** for comparing transit scenarios across a nine-stop Toronto corridor, connecting Zod-validated APIs on Railway to PostgreSQL storage and Python simulation workers.
- Made interrupted simulation jobs recoverable and prevented stale workers from overwriting newer results using **PostgreSQL job queues, expiring leases, and ownership checks**.
- Automated assumption testing through the **Gemini API**, validating structured outputs, numbers, and citations in code and applying deterministic fallbacks when checks fail.

AI Quant Council (IgnitionHacks V7) | *Python, FastAPI, FAISS* | Code | Live — August 2026

- Built an event-driven backtester to evaluate seven-stock portfolios from a **499-stock universe**, incorporating commissions, volume-based slippage, and position limits.
- Restricted retrieval to filings available at each backtest date using **point-in-time filtering** across a **26,000-chunk SEC corpus** in FAISS, supported by a **229-test pipeline suite**.
- Moved long-running research off the HTTP request path using **background jobs and progress-event streaming**, with versioned disk caching to reuse completed results.

Technical Skills

Languages: Python, C++, TypeScript/JavaScript, Java, C, SQL
Frameworks & Libraries: Node.js, Express, React, FastAPI, Flask, Google ADK, pybind11, Zod, FAISS
Infrastructure & Tools: AWS, GCP, Docker, Linux, Git, GitHub Actions, CMake, PostgreSQL, MySQL, MongoDB